

Taleef Tamsal

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SUMMARY

Software engineer building production AI agents and backend systems in Python and TypeScript. Designed and deployed a containerized multi-service RAG platform for AI agents, migrated a Java/Spring Boot backend to modular TypeScript, and automated large-scale LLM safety evaluations. Skilled in integrating AI models with existing infrastructures and data pipelines.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash, C++, C#
- **AI / ML:** PyTorch, Hugging Face Transformers, LangChain, scikit-learn, Pandas, RAG, Cursor, Claude Code, Codex
- **Backend & Web:** Node.js, Express, FastAPI, Spring Boot, Next.js, React, REST APIs, OpenSearch, Redis, PostgreSQL
- **Cloud & DevOps:** Docker, GitHub Actions, CI/CD, Azure, AWS, Linux, GCP, HPC, SLURM
- **Certifications:** DeepLearning.AI Data Engineering, Databricks Data Engineer Associate

EXPERIENCE

Medical Informatics Engineering

May 2026 - Present

Software Developer

- Led development of WorkWell Measure Studio, migrating the backend from Java 21/Spring Boot to modular TypeScript while preserving REST API contracts and deterministic CQL behavior; deployed Next.js 16, PostgreSQL, Docker, and GitHub Actions on MIE infrastructure.
- Implemented WebChart interoperability using HL7 FHIR R4 and SMART Backend Services, plus FHIR MeasureReport, MAT-compatible, and QRDA exports; validated CMS122 and CMS125 against 121/121 official MADiE test cases.
- Scaled audited batch evaluation to 5,000 subjects and 72,100 outcomes using bounded-memory, resumable data pipelines with per-subject failure isolation.
- Implemented tool calling and agent orchestration capabilities by adding guardrailed AI and read-only MCP tools for CQL drafting, test-fixture generation, evidence explanation, and agent access while keeping deterministic CQL as the sole compliance authority.

Purdue University

Aug 2025 - May 2026

Graduate Researcher - AI Safety & LLM Evaluation

- Automated FSDP and SLURM data pipelines to fine-tune 25 LLM configurations across five model families using full fine-tuning, LoRA, QLoRA, FP16, and 4-bit quantization on A100/A30 GPUs, enabling systematic comparison of model families and reducing tuning runtime.
- Executed PAIR, DeepInception, ArtPrompt, and AutoDAN across HarmBench and JailbreakBench, producing nearly 600 attack-success-rate measurements across thousands of completions and analyzing over-refusal and safety-utility trade-offs.
- Calibrated three automated safety evaluators against human annotations and applied bootstrap confidence intervals, nonparametric hypothesis tests, and interaction analysis; open-sourced the experimental pipeline and defended the thesis at the PFW-ETCS Innovation Summit.
- Published two first-author SemEval 2026 system papers on multi-turn RAG and political-response clarity/evasion classification in the ACL Anthology.

Medical Informatics Engineering

May 2025 - Sep 2025

Software Development Intern

- Architected a containerized, multi-service RAG platform using Node.js, Docker, OpenSearch, Redis, and PostgreSQL, decoupling asynchronous document ingestion from query serving for up to 25 concurrent users and AI agents at sub-500 ms retrieval latency.
- Built a retrieval pipeline combining HyDE query expansion, hybrid search, parent-child retrieval, deduplication, reranking, and feedback boosting; indexed 3,000+ Redmine issues and achieved >85% context relevance measured with TruLens.
- Implemented REST, MCP, and SSE interfaces with structured citations, JWT/API-key authentication, user-scoped security filters, and OpenTelemetry-based observability, which improved system monitoring and enabled quicker detection of performance issues.

PROJECTS

JobOps Copilot - Deployed AI Agent Platform [GitHub](#)

2026

- Built and deployed an AI job-operations CRM with a Next.js 16/React 19 frontend, Express/TypeScript API, Python FastAPI agent service, Azure App Service, Azure PostgreSQL, Clerk authentication, and n8n workflow automation.
- Developed LangChain agents for job tracking, company research, interview preparation, skill-gap analysis, and outreach drafting, using multi-provider LLM routing, pgvector-based RAG, tool calling, and human-in-the-loop controls.
- Monitored and evaluated agent performance using real customer data, refining prompt strategies and data pipelines to improve response accuracy and system reliability.

TurboQuant - KV-Cache Compression for Long-Context LLMs [GitHub](#)

2026

- Reimplemented TurboQuant-style KV-cache compression in PyTorch, integrating rotation, Lloyd-Max quantization, Hugging Face DynamicCache, and paired long-context evaluation.
- Achieved 5.2× KV-cache compression while matching baseline completion quality in tested suites; maintained Qwen2.5-7B retrieval within 1.39 pp of baseline at 32K context.

EDUCATION

Purdue University

Aug 2024 - May 2026

M.S., Computer Science

- GPA: 4.0

Lahore University of Management Sciences

Aug 2020 - May 2024

B.S., Computer Science